

4 (a) Explain what is meant by the *internal energy* of a gas.

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.....
..... [1]

(b) An ideal gas has a volume of $1.8 \times 10^{-3} \text{ m}^3$ at a pressure of $3.3 \times 10^5 \text{ Pa}$ and a temperature of 37°C .

(i) Show that the number of molecules in the gas is 1.4×10^{23} .

[1]

(ii) The gas expands so that its volume increases to $2.4 \times 10^{-3} \text{ m}^3$ at a pressure of $2.3 \times 10^5 \text{ Pa}$ and a temperature of 15°C . The work done by the gas during the expansion is 85 J.

Calculate

1. the change in internal energy of the gas,

change = J [2]

2. the thermal energy required by the gas,

energy = J [2]

3. the ratio

$$\frac{\text{r.m.s. speed of the gas molecules at } 15^{\circ}\text{C}}{\text{r.m.s. speed of the gas molecules at } 37^{\circ}\text{C}} .$$

ratio = [2]

5 (a) Explain what is meant by a progressive transverse wave